Tree

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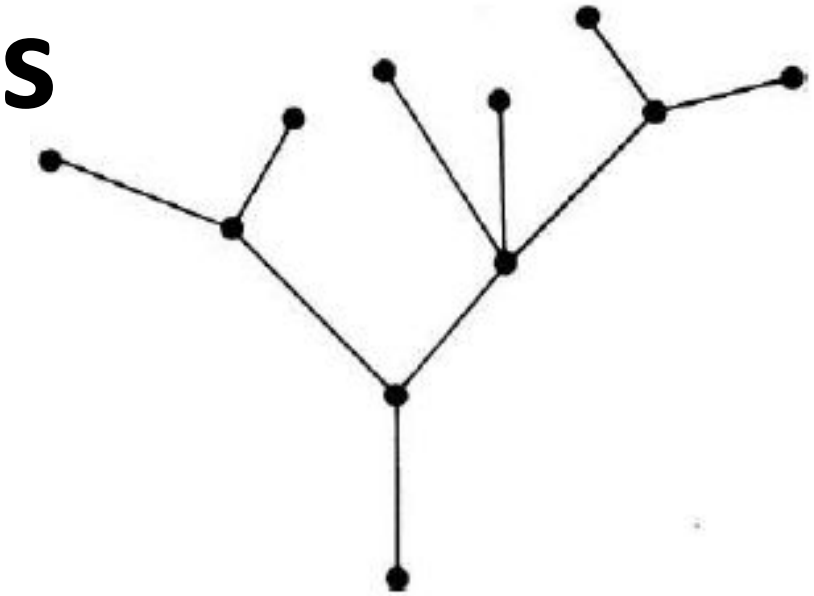
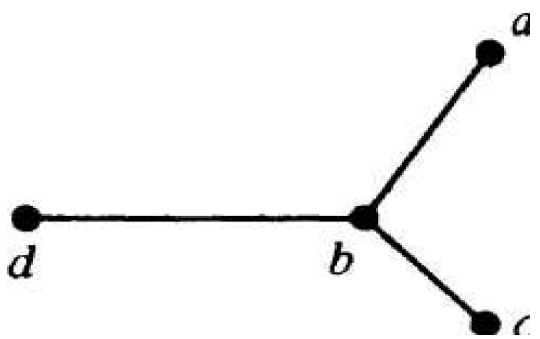
Tree definitions



- A **graph G** with **n vertices** is **called** a **Tree** if
  1. **G** is **connected and** is **circuit less**, or
  2. **G** is **connected and** has **n – 1** edges, or
  3. **G** is **circuit less and** has **n – 1** edges, or
  4. There is **exactly one path between every pair of vertices** in G, or
  5. **G** is **minimally connected** (removal of any one edge makes the graph disconnected).
- **Pendant vertices** in a tree is **defined** as **a vertex of degree one**.
- **In any tree (with two or more vertices), there are at least two pendant vertices.**
  - **In a tree** of **n (n>=2) vertices**, we have **n – 1 edges**, and hence **the total degree 2(n – 1)** is **distributed among n vertices. Since no vertex can be of zero degree, there must be at least two vertices** of **degree one** (**pendant vertices**) **in a tree**.

Eccentricity

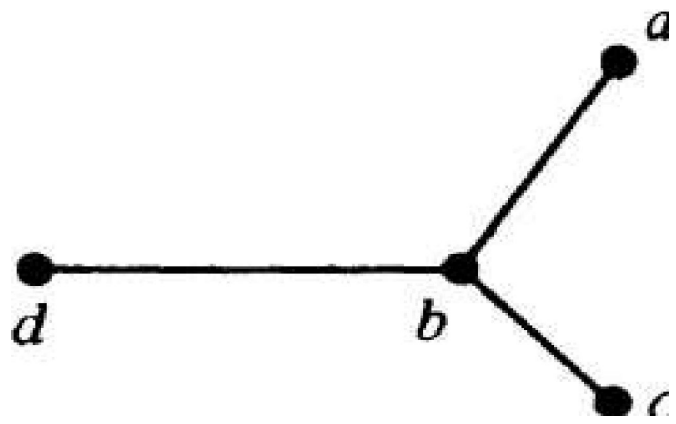
- A **function** that satisfies **following three conditions** is called a ***metric***
  1. **Nonnegativity**: $f(x, y) \geq 0$, and $f(x, y) = 0$ if and only if $x = y$.
  2. **Symmetry**: $f(x, y) = f(x, y)$.
  3. **Triangle inequality**: $f(x, y) \leq f(x, z) + f(z, y)$ for any z.
- In a **connected graph**, the **distance** $d(v_i, v_j)$, is a **metric**. Here $d(v_i, v_j)$ is the **length of the shortest path** (i.e., the **number of edges in the shortest path**) between vertices v_i and v_j.
- The **eccentricity** $E(v)$ of a **vertex** v in a **graph** G is the **distance from** v to the **vertex farthest from** v in G; that is
  - $E(v) = \max_{v_i \in G} d(v, v_i)$
- In **tree G**, $E(a) = 2$, $E(b) = 1$, $E(c) = 2$, and $E(d) = 2$.



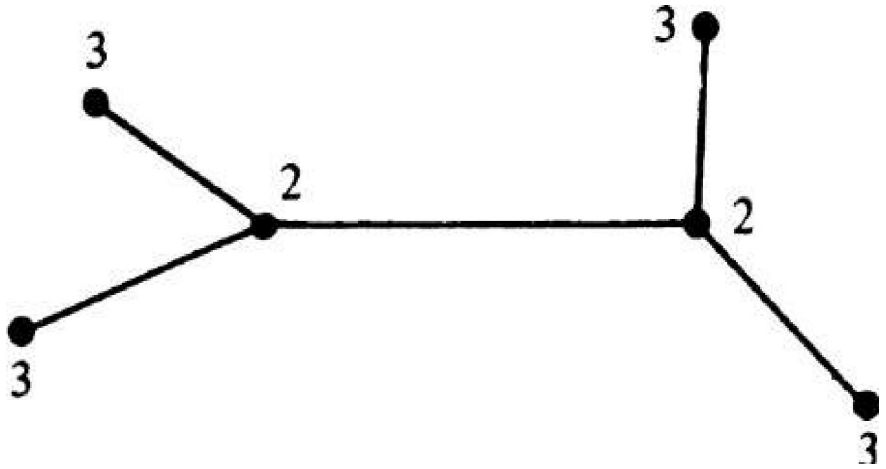
tree G

Eccentricity

- A **vertex** **with** **minimum eccentricity** **in graph** ***G*** is called a ***center*** **of** ***G***.
- As in **tree G**, $E(a) = 2$, $E(b) = 1$, $E(c) = 2$, and $E(d) = 2$, the **vertex** ***b*** is the **center of the tree**.
- **In tree H**, eccentricity of the each of the six vertices is shown next to the vertex. This tree has **two vertices** **having** the same **minimum eccentricity 2**. Hence this tree has **two centers** and is referred to such centers as ***bi-centers***.
- **Every tree** has **either one or two centers**. A **graph**, in general, **has many centers**.
- The **eccentricity of a center** in a tree is defined as the **radius** **of the tree**.



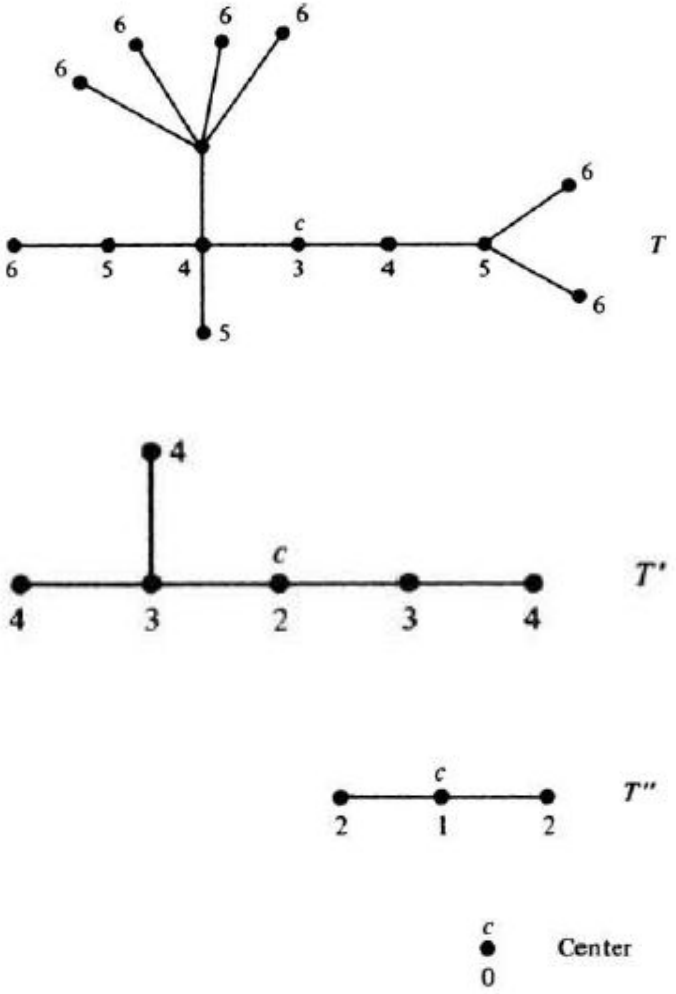
graph G



graph H

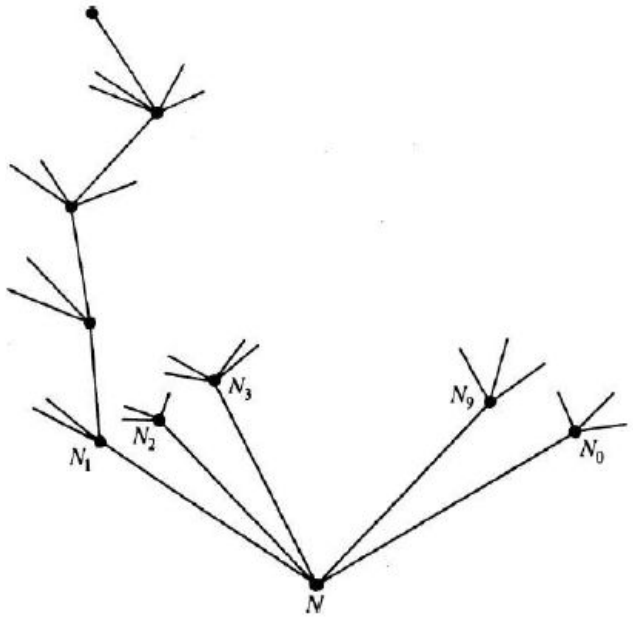
Eccentricity

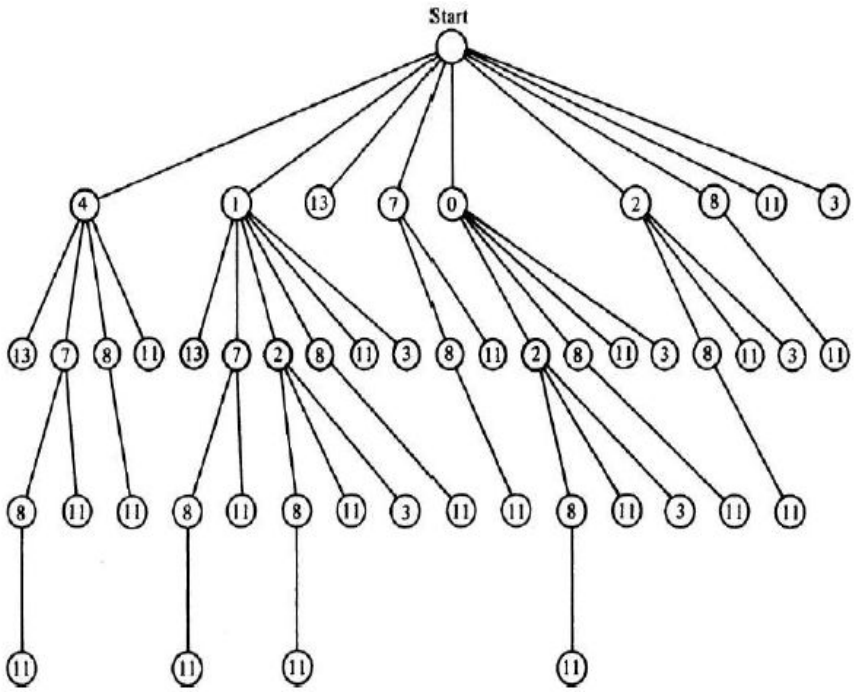
- **Theorem**: **Every tree has either one or two centers.**
- **Proof**: The **maximum distance**, **max $d(v, v_i)$**, from a given **vertex** v to any other **vertex** v_i occurs **only when** v_i **is a pendant vertex**.
- With this observation, let us start with **a tree T having more than two vertices**. Tree T must **have two or more pendant vertices**. **Delete all the pendant vertices** from T.
- The **resulting graph T'** is still **a tree**. Removal of all pendant vertices from T **uniformly reduced the eccentricities of the remaining vertices** (i.e., vertices in T') **by one**. Therefore, **all vertices that T had as centers** will **still remain centers in T'**.
- Continuing this process (as illustrated in Fig) until there is left either a vertex (which is the center of *T*) or an edge (whose end vertices are the two centers of *T*). Thus the theorem.
- **Theorem**: **if a tree T has two centers, the two centers must be adjacent**.
- **Proof**: The **two centers if exists**, they are the **endpoints of an edge**. Hence proved.

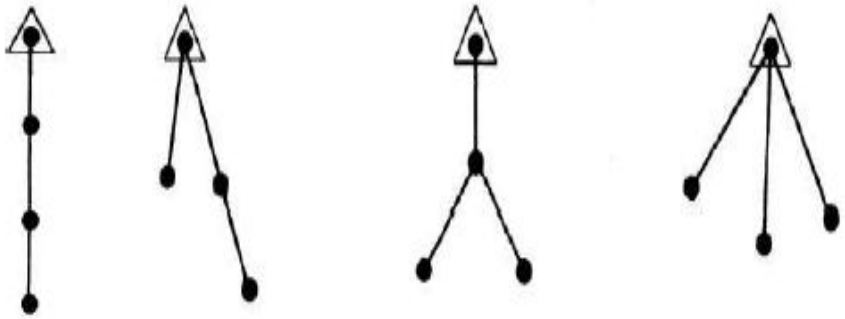


Rooted Tree

- A **Tree** can be **without a root** **or** **with a root**.
- A tree in which **one vertex** (called the **root**) **is distinguished** from all the others is called a **rooted tree**. Nodes such as **N** and **start** are the **root** of the trees in the graphs shown below.
- In a rooted tree, the **root** is generally **marked** distinctly and may be enclosed in **a small triangle**.



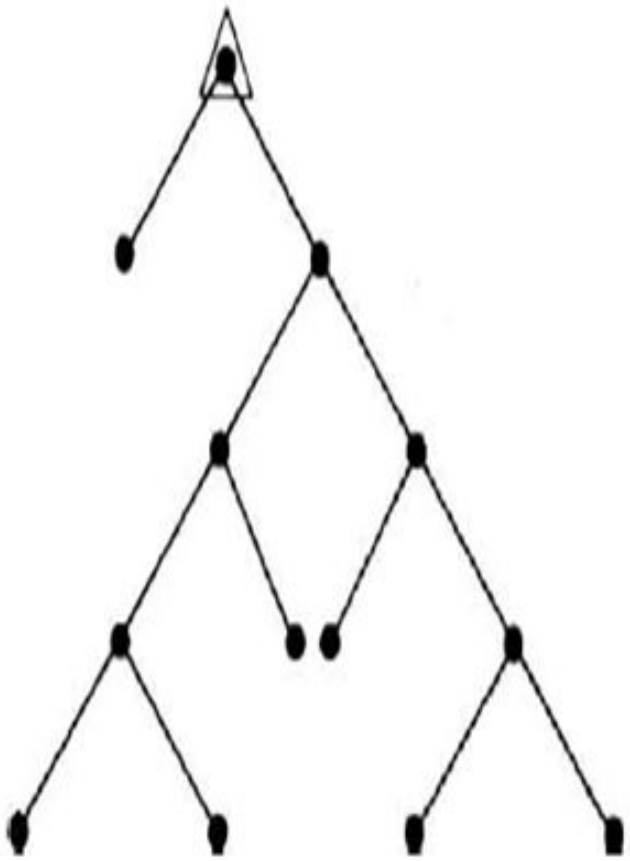




Rooted trees with four vertices

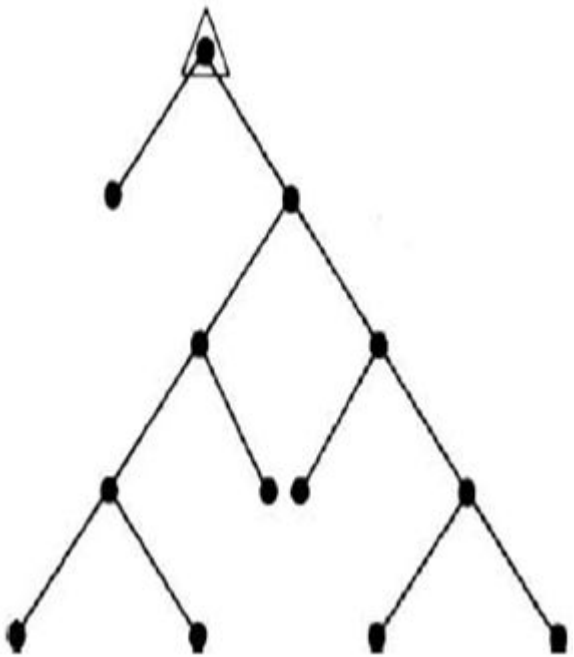
Binary Tree

- If **binary tree** is defined as a **tree** in which there is **exactly one vertex** of **degree two**, and the **remaining vertices** are of **degree one or three** (as shown in Fig). Here the **trees** are with **three** or **more vertices**.
- The **vertex of degree two** is distinct from all other vertices, this vertex serves as a **root**.
- Every **binary tree** is a **rooted tree**.
- The **number of vertices n in a binary tree is always odd** (for **trees** with ***n* >= 3**). **This is because there is exactly one vertex** (root) of **even degree**, and the **remaining n – 1 vertices** are of **odd degrees**. **Since** the **number of vertices of odd degrees** is **even**, **n – 1** is **even**. Hence **n** is **odd**.
- **The number of edges** in a **binary tree with n vertices** equals to **n-1**.



Binary Tree

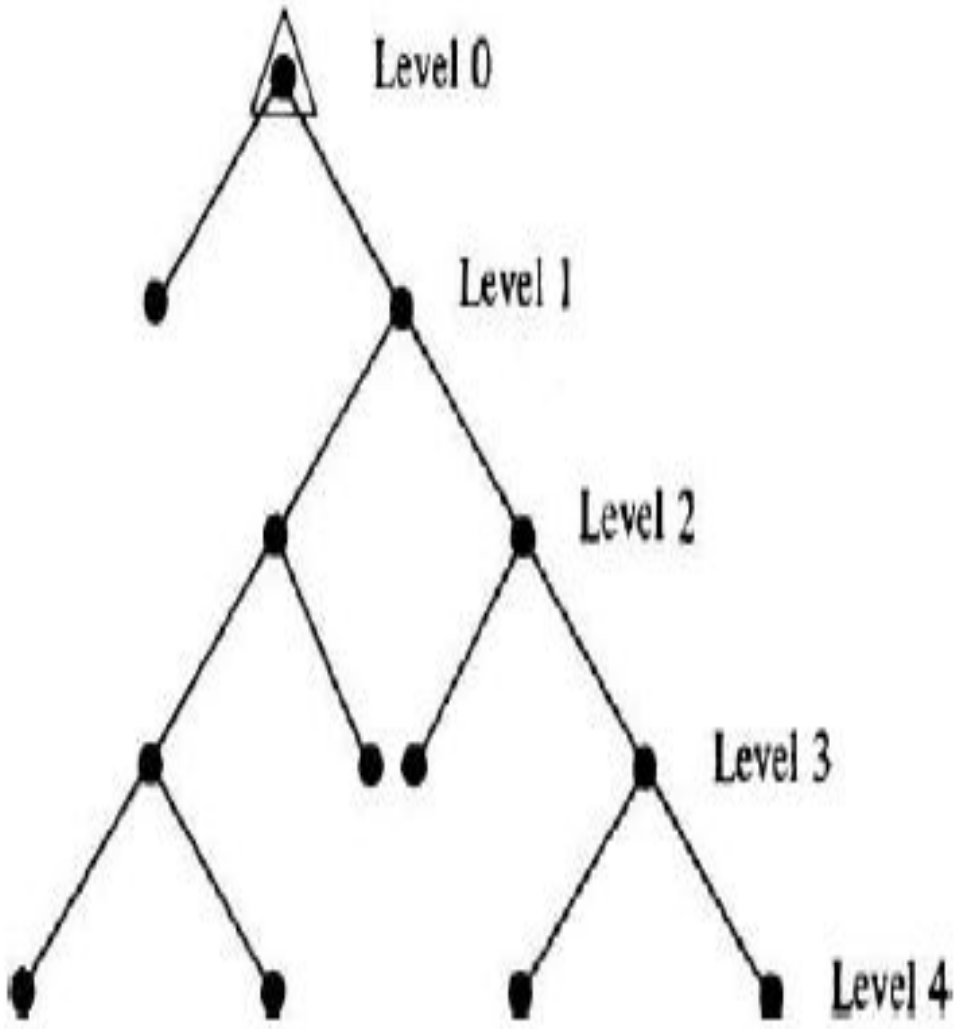
- The number of **pendant vertices** in a **binary tree** of **n nodes** is **p = (n + 1)/2**
- Proof:
  1. Let **p** is the **number of pendant vertices** in a binary tree **T**. *T*hen **n − p − 1** is the **number of vertices of degree three**. [Out of **n** nodes, one node root is with degree 2]. Therefore, the **total number of edges** in **T** can be **[p + 3(n – p – 1) + 2]/2**.
  2. In a binary tree of **n** nodes, the **total number of edges** is known to be equal to **n – 1**.
     - Equating the above two, we can show that the **number of pendant vertices p = (n + 1)/2**
- As the **total number of nodes** is **n**, the **number of internal** (**non-pendent**) nodes is **n – (n + 1)/2** = (**n – 1)/2**. Therefore, **number of internal nodes** is **one less than the number of pendent nodes**.



No. of nodes: 13
No. of edges: 12
No. of internal nodes: 6
No. of pendent nodes: 7

Level of nodes in a Binary Tree

- In a **binary tree**, a **vertex** v_i is said to be at **level** l_i **if** v_i is at a **distance** of l_i **from the root**. (distance is no. of edges)
- Thus the **root** is at **level 0**.
- **Maximum number of vertices** in the level **i** is 2^i.
- A 13-vertex, **four-level** binary tree is shown in **Fig**. The number of vertices at levels 1, 2, 3, and 4 are 2, 2, 4, and 4, respectively.
- The **maximum level**, l_{max}, of any vertex in a binary tree is called the **height** of the tree.



Level of nodes in a Binary Tree

- The **minimum possible height** of an **n-vertex binary tree** is

  $\min l_{max} = \lceil \log_2 (n+1) - 1 \rceil$,

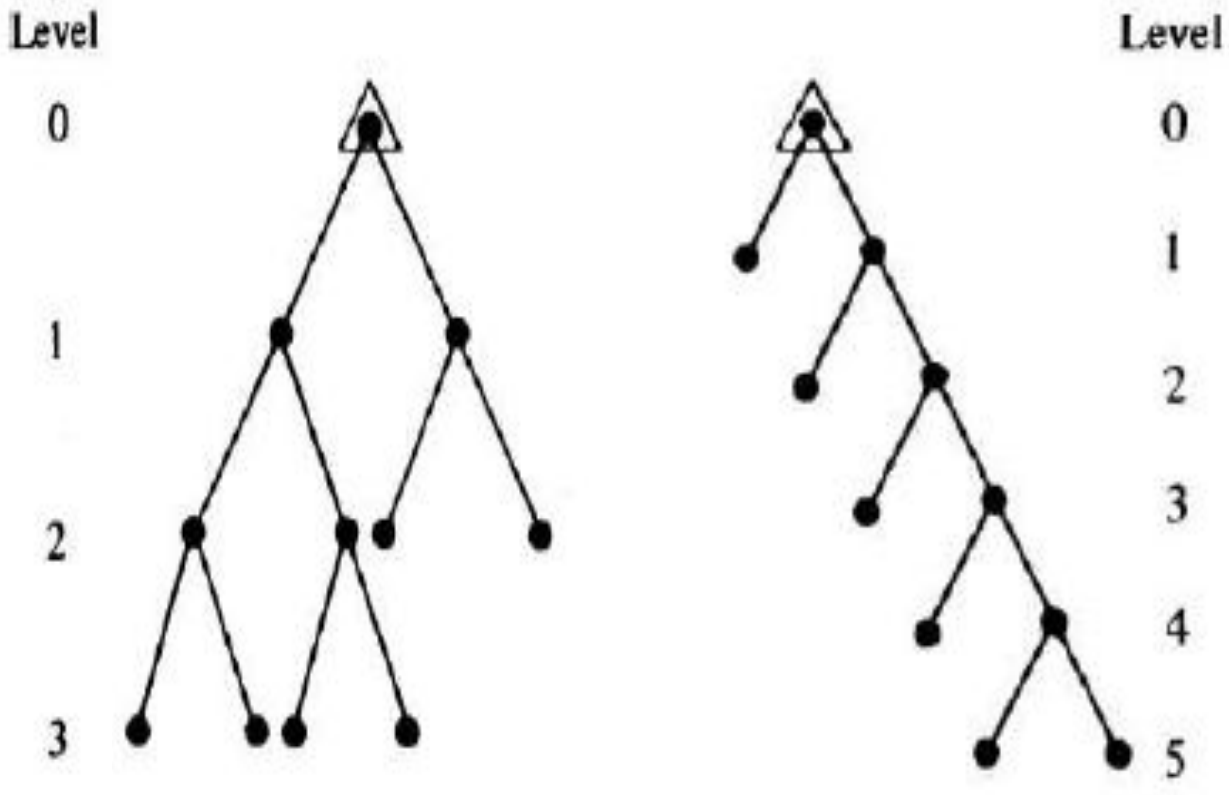
Here $\lceil n \rceil$ denotes the **smallest integer greater than or equal to n**.

- **maximum possible height** of an **n-vertex binary tree** is

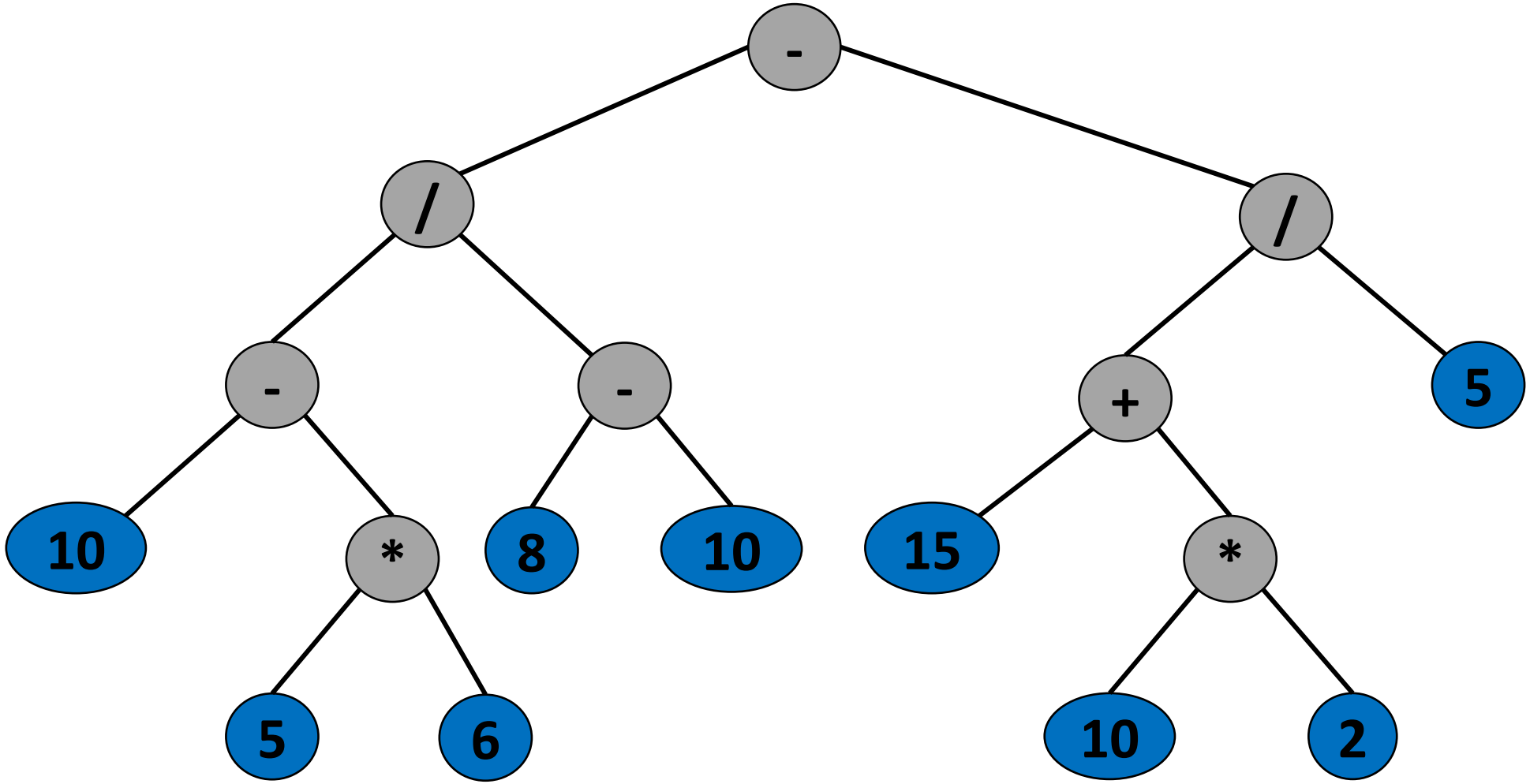
  $\max l_{max} = (n - 1)/2$

Maximum possible **height** can be **achieved**, by constructing a binary tree for a given **n** such that the **farthest vertex is as far as possible from the root**. For this, we must have **exactly two vertices at each level, except at the 0 level**.

- For **n = 11**, **binary trees** realizing **both these extreme heights** are **shown** in **Fig**



$\min l_{max} = \lceil \log_2 (11+1) - 1 \rceil = 3$ $\max l_{max} = (11 - 1)/2 = 5$

Arithmetic Expression in a Binary Tree

(10 – 5 * 6)/(8 – 10) – (15 + 10 * 2)/5